Linear Burger equation : $\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = \nu\frac{\partial^2 u}{\partial x^2}$

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = (1-\theta)\left(\frac{\partial u}{\partial t}\right)_i^n + \theta\left(\frac{\partial u}{\partial t}\right)_i^{n+1}$$

For consistency : $\lim_{\Delta x, \Delta t \to 0} (FDE - PDE)_i^n$

How to check ?

→ Expand terms via Taylor & then substitute

$$u_i^{n+1} = u_i^n + \left(\frac{\partial u}{\partial t}\right)_i^n \Delta t + \left(\frac{\partial^2 u}{\partial t^2}\right)_i^n \frac{\Delta t^2}{2!} + \left(\frac{\partial^3 u}{\partial t^3}\right)_i^n \frac{\Delta t^3}{3!} + \cdots$$

$$u_{i-1}^n = u_i^n - \left(\frac{\partial u}{\partial x}\right)_i^n \Delta x + \left(\frac{\partial^2 u}{\partial x^2}\right)_i^n \frac{\Delta x^2}{2!} + \left(\frac{\partial^3 u}{\partial x^3}\right)_i^n \frac{\Delta x^2}{3!} + \cdots$$

$$u_{i+1}^n = u_i^n + \left(\frac{\partial u}{\partial x}\right)_i^n \Delta x + \left(\frac{\partial^2 u}{\partial x^2}\right)_i^n \frac{\Delta x^2}{2!} + \left(\frac{\partial^3 u}{\partial t^3}\right)_i^n \frac{\Delta x^3}{3!} + \cdots$$

$$u_{i+1}^{n+1} = u_i^n + \left(\frac{\partial u}{\partial t}\right)_i^n \Delta t + \left(\frac{\partial u}{\partial x}\right)_i^n \Delta x + \frac{1}{2!}\left[\left(\frac{\partial^2 u}{\partial t^2}\right)_i^n \Delta t^2 + \left(\frac{\partial^2 u}{\partial x^2}\right)_i^n \Delta x^2 + \frac{\partial^2 u}{\partial x \partial t}\Delta x \Delta t\right] + \frac{1}{3!}\left[\left(\frac{\partial^3 u}{\partial t^3}\right)_i^n \Delta t^3 + 3\left(\frac{\partial^3 u}{\partial t^2 \partial x}\right)_i^n \Delta t^2 \Delta x + 3\left(\frac{\partial^3 u}{\partial t \partial x^2}\right)_i^n \Delta t \Delta x^2 + \left(\frac{\partial^3 u}{\partial x^3}\right)_i^n \Delta x^3\right] + \cdots$$

$$u_{i-1}^{n+1} = u_i^n + \left(\frac{\partial u}{\partial t}\right)_i^n \Delta t - \left(\frac{\partial u}{\partial x}\right)_i^n \Delta x + \frac{1}{2!}\left[\left(\frac{\partial^2 u}{\partial t^2}\right)_i^n \Delta t^2 + \left(\frac{\partial^2 u}{\partial x^2}\right)_i^n \Delta x^2 - \frac{\partial^2 u}{\partial x \partial t}\Delta x \Delta t\right] + \frac{1}{3!}\left[\left(\frac{\partial^3 u}{\partial t^3}\right)_i^n \Delta t^3 - 3\left(\frac{\partial^3 u}{\partial t^2 \partial x}\right)_i^n \Delta t^2 \Delta x + 3\left(\frac{\partial^3 u}{\partial t \partial x^2}\right)_i^n \Delta t \Delta x^2 - \left(\frac{\partial^3 u}{\partial x^3}\right)_i^n \Delta x^3\right] + \cdots$$

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = (1-\theta)\left(-c\frac{u_{i+1}^n - u_{i-1}^n}{2\Delta x} + \nu\frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2}\right) + \theta\left(-c\frac{u_{i+1}^{n+1} - u_{i-1}^{n+1}}{2\Delta x} + \nu\frac{u_{i+1}^{n+1} - 2u_i^{n+1} + u_{i-1}^{n+1}}{\Delta x^2}\right)$$

$\left(\frac{\partial u}{\partial t}\right)_i^n \Delta t$; $\left(\frac{\partial u}{\partial x}\right)_i^n \Delta x + \left(\frac{\partial^3 u}{\partial x^3}\right)_i^n \frac{\Delta x^3}{3!} \cdots$; $\left(\frac{\partial^2 u}{\partial x^2}\right)_i^n \frac{\Delta x^2}{2!} + \left(\frac{\partial^4 u}{\partial x^4}\right)_i^n \frac{\Delta x^4}{4!} \cdots$

doing this for all terms

& getting $\Delta t \to 0$, $\Delta x \to 0$

Question 2

Fourier method of stability analysis (disturbance at every grid) $\varepsilon_i^n = \sum_{-N}^{N} A_j^n e^{I k_j x_i}$

↓

$$\frac{(u_i + \varepsilon)^{n+1} - (u_i + \varepsilon)^n}{\Delta t} = (1-\theta)\left[\cdots\right.$$

Then

$$\varepsilon_i^{n+1} = \sum A_j^{n+1} \exp\left(I k_j x_i\right)$$

$$\varepsilon_i^n = \sum A_j^n \exp\left(I k_j x_i\right)$$

$$\varepsilon_{i+1}^n = \sum A_j^n e^{I k_j x_{i+1}} \qquad x_{i+1} = x_i + \Delta x \qquad x_{i-1} = x_i - \Delta x$$

$\sin\left(I k_j x_{i+1}\right) + \cos\left(I k_j x_{i+1}\right)$

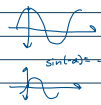
$\cos(-\alpha) = \cos\alpha$

$\exp\left(I k_j \Delta x\right) - \exp\left(-I k_j \Delta x\right) = \sin\left(I k_j x\right)$

$\sin(-\alpha) = -\sin\alpha$

$i\sin(\alpha) + \cos\alpha - \left(i\sin(-\alpha) + \cos(-\alpha)\right)$

$i\sin(\alpha) + \cos\alpha + \sin\alpha - \cos(\alpha)$



$\exp(-I\omega_j \Delta x) - \exp(-\omega_j \Delta x) = \sin(\omega_j x)$

$i\sin(\alpha) + \cos\alpha - (i\sin(-\alpha) + \cos(-\alpha))$

$i\sin(\alpha) + \cos\alpha + \sin\alpha - \cos(\alpha)$

↳ Use this analysis:

expression for A_j^n

↓

define G

↓

find $|G|^2 \rightarrow |x|^2 = x$ (modulus)

$z = x + iy$

$|z|^2 = x^2 + y^2$

$|G| \leq 1$

$|G|^2 \leq 1$

Lax Convergence Theorem

↳ If FDE/FVE's are well-posed, linear PDE's are convergent if the FDE's/FVE's are consistent & stable

↓

→ solution exists & is unique

→ solution depends cont. on the data

Question 1

If $\theta=1$, will FD/FV yield exact solution as $\Delta x, \Delta t \to 0$

Modified equation becomes, $(\theta=1)$:

$$\frac{u_i^{n+1}-u_i^n}{\Delta t} = -c\left(\frac{u_{i+1}^{n+1}-u_{i-1}^{n+1}}{2\Delta x}\right) + \nu\,\frac{\left(u_{i+1}^{n+1}-2u_i^{n+1}+u_{i-1}^{n+1}\right)}{\Delta x^2}$$

To yield exact soln as $\Delta x, \Delta t \to 0$; have to check for convergence (global error goes to 0 as $\Delta x, \Delta t \to 0$)

$$\lim_{\Delta x,\,\Delta t\to 0} N_i^n = U_i^n \quad \text{, where}$$

N_i^n = numerical solution @ (i,n) w/ round off errors

U_i^n = exact solution @ (i,n)

According to Equivalence theorem of Lax,

FDE will be convergent for well-posed, linear PDE's if the FDE is consistent & stable

Showing consistency:

→ Expand terms via Taylor & then substitute

$$u_i^{n+1} = u_i^n + \left(\frac{\partial u}{\partial t}\right)_i^n \Delta t + \left(\frac{\partial^2 u}{\partial t^2}\right)_i^n \frac{\Delta t^2}{2!} + \left(\frac{\partial^3 u}{\partial t^3}\right)_i^n \frac{\Delta t^3}{3!} + \cdots$$

$$u_{i-1}^n = u_i^n - \left(\frac{\partial u}{\partial x}\right)_i^n \Delta x + \left(\frac{\partial^2 u}{\partial x^2}\right)_i^n \frac{\Delta x^2}{2!} + \left(\frac{\partial^3 u}{\partial x^3}\right)_i^n \frac{\Delta x^3}{3!} + \cdots$$

$$u_{i+1}^n = u_i^n + \left(\frac{\partial u}{\partial x}\right)_i^n \Delta x + \left(\frac{\partial^2 u}{\partial x^2}\right)_i^n \frac{\Delta x^2}{2!} + \left(\frac{\partial^3 u}{\partial x^3}\right)_i^n \frac{\Delta x^3}{3!} + \cdots$$

$$u_{i+1}^{n+1} = u_i^n + \left(\frac{\partial u}{\partial t}\right)_i^n \Delta t + \left(\frac{\partial u}{\partial x}\right)_i^n \Delta x + \frac{1}{2!}\left[\left(\frac{\partial^2 u}{\partial t^2}\right)_i^n \Delta t^2 + \left(\frac{\partial^2 u}{\partial x^2}\right)_i^n \Delta x^2 + \frac{\partial^2 u}{\partial x \partial t}\Delta x \Delta t\right] + \frac{1}{3!}\left[\left(\frac{\partial^3 u}{\partial t^3}\right)_i^n \Delta t^3 + 3\left(\frac{\partial^3 u}{\partial t^2 \partial x}\right)_i^n \Delta t^2 \Delta x + 3\left(\frac{\partial^3 u}{\partial t \partial x^2}\right)_i^n \Delta t\,\Delta x^2 + \left(\frac{\partial^3 u}{\partial x^3}\right)_i^n \Delta x^3\right] + \cdots$$

$$u_{i-1}^{n+1} = u_i^n + \left(\frac{\partial u}{\partial t}\right)_i^n \Delta t - \left(\frac{\partial u}{\partial x}\right)_i^n \Delta x + \frac{1}{2!}\left[\left(\frac{\partial^2 u}{\partial t^2}\right)_i^n \Delta t^2 + \left(\frac{\partial^2 u}{\partial x^2}\right)_i^n \Delta x^2 - \frac{\partial^2 u}{\partial x \partial t}\Delta x \Delta t\right] + \frac{1}{3!}\left[\left(\frac{\partial^3 u}{\partial t^3}\right)_i^n \Delta t^3 - 3\left(\frac{\partial^3 u}{\partial t^2 \partial x}\right)_i^n \Delta t^2 \Delta x + 3\left(\frac{\partial^3 u}{\partial t \partial x^2}\right)_i^n \Delta t\,\Delta x^2 - \left(\frac{\partial^3 u}{\partial x^3}\right)_i^n \Delta x^3\right] + \cdots$$

$$\frac{u_i^{n+1}-u_i^n}{\Delta t} = -c\left(\frac{u_{i+1}^{n+1}-u_{i-1}^{n+1}}{2\Delta x}\right) + \nu\,\frac{\left(u_{i+1}^{n+1}-2u_i^{n+1}+u_{i-1}^{n+1}\right)}{\Delta x^2}$$

$$\left[\left(\frac{\partial u}{\partial t}\right)_i^n + \frac{1}{2}\left(\frac{\partial^2 u}{\partial t^2}\right)_i^n \Delta t + \cdots\right] = \frac{-c}{2\Delta x}\left[2\left(\left(\frac{\partial u}{\partial x}\right)_i^n \Delta x\right) + \frac{\partial^2 u}{\partial x \partial t}\Delta x \Delta t + \left(\frac{\partial^3 u}{\partial t^2 \partial x}\right)_i^n \Delta t^2 \Delta x + \frac{1}{3}\left(\frac{\partial^3 u}{\partial x^3}\right)_i^n \Delta x^3\right] + \frac{\nu}{\Delta x^2}\left[\left(2u_i + 2\left(\frac{\partial u}{\partial t}\right)_i^n \Delta t + \left(\frac{\partial^2 u}{\partial t^2}\right)_i^n \Delta t^2 + \left(\frac{\partial^2 u}{\partial x^2}\right)_i^n \Delta x^2 + \frac{1}{3!}\left[2\frac{\partial^3 u}{\partial t^3}\Delta t^3 + 6\frac{\partial^3 u}{\partial t \partial x^2}\Delta t \Delta x^2\right]\right) - 2u_i - 2\left(\frac{\partial u}{\partial t}\right)_i^n \Delta t - 2\left(\frac{\partial^2 u}{\partial t^2}\right)_i^n \frac{\Delta t^2}{2!} - 2\frac{\partial^3 u}{\partial t^3}\frac{\Delta t^3}{3!}\right]$$

$$\therefore \left[\left(\frac{\partial u}{\partial t}\right)_i^n + \frac{1}{2}\left(\frac{\partial^2 u}{\partial t^2}\right)_i^n \Delta t + \cdots\right] = -c\left[\left(\frac{\partial u}{\partial x}\right)_i^n + \frac{\partial^2 u}{\partial x \partial t}\frac{\Delta t}{2} + \frac{\partial^3 u}{\partial t^2 \partial x}\frac{\Delta t^2}{2} + \frac{1}{3}\left(\frac{\partial^3 u}{\partial x^3}\right)_i^n \frac{\Delta x^2}{2}\right] + \nu\left[\left(\frac{\partial^2 u}{\partial x^2}\right)_i^n + 2\left(\frac{\partial^3 u}{\partial t \partial x^2}\right)\Delta t\right]$$

As $\Delta t, \Delta x \to 0$

$$\therefore \left[\left(\frac{\partial u}{\partial t}\right)_i^n + \frac{1}{2}\left(\frac{\partial^2 u}{\partial t^2}\right)_i^n \Delta t + \cdots\right] = -c\left[\left(\frac{\partial u}{\partial x}\right)_i^n + \frac{\partial^2 u}{\partial x \partial t}\frac{\Delta t}{2} + \frac{\partial^3 u}{\partial t^2 \partial x}\frac{\Delta t^2}{2} + \frac{1}{3}\left(\frac{\partial^3 u}{\partial x^3}\right)_i^n \frac{\Delta x^2}{2}\right] + \nu\left[\left(\frac{\partial^2 u}{\partial x^2}\right)_i^n + 2\left(\frac{\partial^3 u}{\partial t \partial x^2}\right)\Delta t \cdots\right]$$

(each of the Δt / Δx terms → 0)

$$\left(\frac{\partial u}{\partial t}\right)_i^n = -c\left(\frac{\partial u}{\partial x}\right)_i^n + \nu\left(\frac{\partial^2 u}{\partial x^2}\right)_i^n$$

This FDE is equal to the PDE & therefore unconditionally consistent (no Δt or Δx terms remaining & hence independent of the grid & time spacing)

Determining stability:

Fourier transform analysis

$$\frac{(u_i+\epsilon_i)^{n+1}-(u_i+\epsilon_i)^n}{\Delta t} = -c\,\frac{\left((u_{i+1}+\epsilon_{i+1})^{n+1}-(u_{i-1}+\epsilon_{i-1})^{n+1}\right)}{2\Delta x} + \nu\,\frac{\left((u_{i+1}+\epsilon_{i+1})^{n+1}-2(u_i+\epsilon_i)^{n+1}+(u_{i-1}+\epsilon_{i-1})^{n+1}\right)}{\Delta x^2}$$

Subtracting the FDE from the one above, we get:

$$\frac{\epsilon_i^{n+1}-\epsilon_i^n}{\Delta t} = -c\,\frac{\left(\epsilon_{i+1}^{n+1}-\epsilon_{i-1}^{n+1}\right)}{2\Delta x} + \frac{\nu}{\Delta x^2}\left(\epsilon_{i+1}^{n+1}-2\epsilon_i^{n+1}+\epsilon_{i-1}^{n+1}\right)$$

$$\epsilon_i^n = \sum_{j=-N}^{N} A_j^n\, e^{(Ikx_i)}$$

$e^{ix} = \cos x + i\sin x$

$e^{ix} + e^{-ix} = \cos x + i\sin x + \cos x - i\sin x = 2\cos x$

$e^{ix} - e^{-ix} = \cos x + i\sin x - \cos x + i\sin x = 2i\sin x$

Removing the summation & looking at terms separately:

$$\frac{1}{\Delta t}\left(A_j^{n+1}\exp(Ikx_i) - A_j^n\exp(Ikx_i)\right) = \frac{-c}{2\Delta x}\left(A_j^{n+1}\exp\left(Ik(x_i+\Delta x)\right) - A_j^{n+1}\exp\left(Ik(x_i-\Delta x)\right)\right) + \frac{\nu}{\Delta x^2}\left(A_j^{n+1}\exp\left(Ik(x_i+\Delta x)\right) - 2A_j^{n+1}\exp(Ikx_i) + A_j^{n+1}\exp\left(Ik(x_i-\Delta x)\right)\right)$$

$$\Rightarrow \frac{\exp(Ikx_i)}{\Delta t}\left(A_j^{n+1}-A_j^n\right) = \frac{-c}{2\Delta x}A_j^{n+1}\exp(Ikx_i)\left(\exp(Ik\Delta x) - \exp(-Ik\Delta x)\right) + \frac{\nu}{\Delta x^2}A_j^{n+1}\exp(Ikx_i)\underbrace{\left[\exp(Ik\Delta x) - 2 + \exp(-Ik\Delta x)\right]}_{2\cos(k\Delta x)-2}$$

$$\Rightarrow \left(A_j^{n+1}-A_j^n\right) = \frac{-c\Delta t}{2\Delta x}A_j^{n+1}2I\sin(k\Delta x) + \frac{\nu\Delta t}{\Delta x^2}A_j^{n+1}2\left[\cos(k\Delta x)-1\right]$$

$$C = \frac{c\Delta t}{\Delta x} \quad ; \quad d = \frac{\nu \Delta t}{\Delta x^2} \quad ; \quad G = A_j^{n+1} / A_j^n$$

$$G - 1 = G\left(-IC\sin(k_j\Delta x) - 2d(1-\cos(k_j\Delta x))\right)$$

$$G = \frac{1}{\left[1 + IC\underbrace{\sin(k_j\Delta x)}_{\alpha} + 2d\underbrace{(1-\cos k_j\Delta x)}_{\beta}\right]}$$

$$G = \frac{1}{1 + I\alpha + \beta}$$

$$\frac{1}{[(1+\beta)+I\alpha]} \cdot \frac{(1+\beta)-I\alpha}{(1+\beta)-I\alpha} = \frac{(1+\beta)-I\alpha}{(1+\beta)^2 - \alpha^2}$$

$|G| \le 1$ or $|G|^2 \le 1$

$$|G|^2 = \frac{(1+\beta)^2}{\left((1+\beta)^2-\alpha^2\right)^2} + \frac{\alpha^2}{\left((1+\beta)^2-\alpha^2\right)^2}$$

$$|G|^2 = \frac{(1+\beta)^2 + \alpha^2}{\left((1+\beta)^2-\alpha^2\right)^2} \le 1$$

$$(1+\beta)^2 + \alpha^2 \le \left((1+\beta)^2 - \alpha^2\right)^2$$

$$(1+\beta)^2 + \alpha^2 \le (1+\beta)^4 + \alpha^4 - 2(1+\beta)^2\alpha^2$$

$$-(1+\beta)^4 + (1+\beta)^2(+2\alpha^2+1) - \alpha^4 + \alpha^2 \le 0$$

$$(1+\beta^2)\left[-(1+\beta)^2 + (2\alpha^2+1)\right] - \alpha^2(\alpha^2+1) \le 0$$

$\alpha = C\sin(k_j\Delta x)$

$\beta = (1-\cos(k_j\Delta x)) \ge 0$

Only for certain values of α, C, d will the above inequality be satisfied

↳ Hence conditionally stable

↳ Hence we cannot say if for $\theta = 1$ the given FD will always give an exact solution

Question 2

Dissipation error → analyzed via Fourier Method

$$\left(|G_{PDE}|^n - |G_{FDE}|^n\right) A_j^0$$

↑ disturbance introduced at the zeroth time scale

Finding $|G_{PDE}|$:

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = \nu\frac{\partial^2 u}{\partial x^2} \quad \text{— differentiating}$$

$$\Rightarrow a_j e^{a_j t} e^{Ik_j x} + c\, e^{a_j t} e^{Ik_j x}(Ik_j) = \nu\left((Ik_j)^2 e^{a_j t} e^{Ik_j x}\right)$$

$$\therefore \left(a_j + cIk_j + \nu k_j\right) = 0$$

$$a_j = -\nu k_j - (ck_j)I$$

$$\therefore a = -\nu k_j \quad ; \quad b = (-ck_j)$$

$$G_{PDE} = \frac{u_j(x, t+\Delta t)}{u_j(x,t)} = e^{a_j \Delta t} = e^{(a+Ib)\Delta t} = e^{-\nu k_j^2 \Delta t} \cdot e^{-cIk_j \Delta t}$$

$$|G_{PDE}| = e^{-\nu k_j^2 \Delta t}$$

Finding $|G_{FDE}|$: (for $\theta = 1/2$)

Given FDE, we can use the fourier method to arrive at the following expression for $\theta = \frac{1}{2}$

$$\frac{(u_i+\varepsilon)^{n+1} - (u_i+\varepsilon)^n}{\Delta t} = \frac{1}{2}\left[-\frac{c\left((u_{i+1}+\varepsilon)^n - (u_{i-1}+\varepsilon)^n\right)}{2\Delta x} + \frac{\nu\left((u_{i+1}+\varepsilon)^n - 2(u_i+\varepsilon)^n + (u_{i-1}+\varepsilon)^n\right)}{\Delta x^2}\right]$$

$$+ \frac{1}{2}\left[-\frac{c\left((u_{i+1}+\varepsilon)^{n+1} - (u_{i-1}+\varepsilon)^{n+1}\right)}{2\Delta x} + \frac{\nu\left((u_{i+1}+\varepsilon)^{n+1} - 2(u_i^{n+1}+\varepsilon) + (u_{i-1}+\varepsilon)^{n+1}\right)}{\Delta x^2}\right]$$

Subtracting from the original FDE & using fourier components we get:

$$\left[\varepsilon_i^n = \sum_{j=1}^{M} A_j^n \exp\left(Ik_j x_i\right)\right] \quad ;$$

$$\left(A_j^{n+1} - A_j^n\right)\exp(Ik_j x_i) = \frac{1}{2}\left[-I\frac{c\Delta t}{\Delta x}A_j^n \exp(Ik_j x_i)\sin(k_j\Delta x) - 2\frac{\nu\Delta t}{\Delta x^2}A_j^n \exp\left[Ik_j x_i\right]\left[1-\cos(k_j\Delta x)\right]\right]$$

$$+ \frac{1}{2}\left[-I\frac{c\Delta t}{\Delta x}A_j^{n+1}\exp(Ik_j x_i)\sin(k_j\Delta x) - 2\frac{\nu\Delta t}{\Delta x^2}A_j^{n+1}\exp\left(Ik_j x_i\right)\left[1-\cos(k_j\Delta x)\right]\right]$$

Defining $\frac{c\Delta t}{\Delta x} = C$ & $\frac{\nu\Delta t}{\Delta x^2} = d$ & $G = \frac{A_j^{n+1}}{A_j^n}$,

$$G - 1 = \frac{1}{2}\left[-IC\sin(k_j\Delta x) - 2d(1-\cos(k_j\Delta x))\right] + \frac{1}{2}G\left[-IC\sin(k_j\Delta x) - 2d(1-\cos(k_j\Delta x))\right]$$

$$\therefore G = \frac{1 - \frac{1}{2}\left[2d(1-\cos(k_j\Delta x)) + IC\sin(k_j\Delta x)\right]}{1 + \frac{1}{2}\left[2d(1-\cos(k_j\Delta x)) + IC\sin(k_j\Delta x)\right]}$$

$$G = \frac{\left[1 - \underbrace{d(1-\cos k_j\Delta x)}_{a}\right] + \underbrace{IC\sin(k_j\Delta x)}_{Im}}{\underbrace{1 + d(1-\cos k_j\Delta x)}_{b} + IC\sin(k_j\Delta x)}$$

$$= \frac{a + Im}{b + Im}$$

$$= \frac{(a+Im)(b-Im)}{b^2+m^2} = \frac{ab - a(Im) + b(Im) + m^2}{b^2+m^2}$$

$$= \frac{ab + Im(b-a) + m^2}{b^2+m^2}$$

$$G = \frac{ab+m^2}{(b^2+m^2)} + I\left(\frac{m(b-a)}{(b^2+m^2)}\right) \qquad ; \quad |z| = \sqrt{x^2+y^2} \quad z = x+iy$$

$$|G|^2 = \left(\frac{ab+m^2}{(b^2+m^2)}\right)^2 + \left(\frac{m(b-a)}{(b^2+m^2)}\right)^2$$

$$|G_{FDE}| = |G| = \left[\frac{(ab+m^2)^2 + m^2(b-a)^2}{(b^2+m^2)^2}\right]^{1/2}$$

$\therefore$ For an initial disturbance of $A_0 = 1$

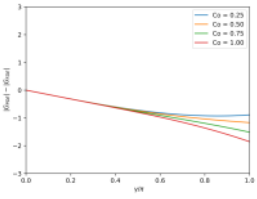
$$(|G_{PDE}|^n - |G_{FDE}|^n) = e^{-\nu k_j^2 \Delta t} - \left[\frac{(ab+m^2)^2 + m^2(b-a)^2}{(b^2+m^2)^2}\right]^{1/2}$$

Plot below for varying C values to see how the above absolute error behaves

1) Error increases quickly as C increases

2) Fourier components w/ the smallest wave no. (longest wavelength) are dissipated the least

3) Stable numerical solutions For smaller Co numbers



Question 3

Dispersion error

Finding ϕ_{PDE} first:

$$u_j(x,t) = e^{\alpha_j t} e^{Ik_j x} \qquad ; \quad \alpha_j = a + Ib$$

$$\left.\begin{aligned} \frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} &= \nu \frac{\partial^2 u}{\partial x^2} \end{aligned}\right\} \text{ differentiating}$$

$$\Rightarrow \alpha_j e^{\alpha_j t} e^{Ik_j x} + c e^{\alpha_j t} e^{Ik_j x} (Ik_j) = \nu\left((Ik_j)^2 e^{\alpha_j t} e^{Ik_j x}\right)$$

$$= (\alpha_j + cIk_j + \nu k_j) = 0$$

$$\alpha_j = -\nu k_j - (ck_j)I$$

$$\therefore a = -\nu k_j \qquad ; \quad b = (-ck_j)$$

$$G_{PDE} = \frac{u_j(x,t+\Delta t)}{u_j(x,t)} = e^{\alpha_j \Delta t} = e^{(a+ib)\Delta t} = e^{-\nu k_j^2 \Delta t} \cdot e^{-ck_j I \Delta t}$$

$$\phi_{PDE} = -ck_j \Delta t$$

Given FDE, we can use the fourier method to arrive at the following expression for $\theta = \frac{1}{2}$

$$\frac{(u_i+\varepsilon)^{n+1} - (u_i+\varepsilon)^n}{\Delta t} = \frac{1}{2}\left[\frac{-c(u_{i+1}+\varepsilon)^n - (u_{i-1}+\varepsilon)^n}{2\Delta x} + \frac{\nu(u_{i+1}+\varepsilon)^n - 2(u_i+\varepsilon)^n + (u_{i-1}+\varepsilon)^n}{\Delta x^2}\right]$$

$$+ \frac{1}{2}\left[\frac{-c(u_{i+1}+\varepsilon)^{n+1} - (u_{i-1}+\varepsilon)^{n+1}}{2\Delta x} + \frac{\nu(u_{i+1}+\varepsilon)^{n+1} - 2(u_i^{n+1}+\varepsilon) + (u_{i-1}+\varepsilon)^{n+1}}{\Delta x^2}\right]$$

Subtracting from the original FDE & using fourier components we get:

$$\left[\varepsilon_i^n = \sum_{j=-N}^{N} A_j^n \exp\left[Ik_j x_i\right]\right] \qquad ;$$

$$(A_j^{n+1} - A_j^n)\exp(Ik_j x_i) = \frac{1}{2}\left[-I\frac{c\Delta t}{\Delta x} A_j^n \exp(Ik_j x_i)\sin(k_j\Delta x) - 2\frac{\nu\Delta t}{\Delta x^2} A_j^n \exp\left[Ik_j x_i\right)\left[1-\cos(k_j\Delta x)\right]\right]$$

$$+ \frac{1}{2}\left[-I\frac{c\Delta t}{\Delta x} A_j^{n+1}\exp(Ik_j x_i)\sin(k_j\Delta x) - 2\frac{\nu\Delta t}{\Delta x^2} A_j^{n+1}\exp\left(Ik_j x_i\right)\left[1-\cos(k_j\Delta x)\right)\right]$$

Defining $\frac{c\Delta t}{\Delta x} = C$ & $\frac{\nu\Delta t}{\Delta x^2} = d$ & $G = \frac{A_j^{n+1}}{A_j^n}$,

$$G - 1 = \frac{1}{2}\left[-IC\sin(k_j\Delta x) - 2d(1-\cos(k_j\Delta x))\right] + \frac{1}{2}G\left[-IC\sin(k_j\Delta x) - 2d(1-\cos(k_j\Delta x))\right]$$

$$\therefore \quad G = \frac{1 - \frac{1}{2}\left[2d(1-\cos(k_j\Delta x)) + IC\sin(k_j\Delta x)\right]}{1 + \frac{1}{2}\left[2d(1-\cos(k_j\Delta x)) + IC\sin(k_j\Delta x)\right]}$$

$$G = \left[\overbrace{1 - \tfrac{1}{2}(\ldots)}^{a}\right] - I\overbrace{C\sin(k_j\Delta x)}^{m}$$

$$\therefore \quad G = \frac{1 - \frac{1}{2}\left[2d(1-\cos(k_j\Delta x)\right] + IC\sin(k_j\Delta x)}{1 + \frac{1}{2}\left[2d(1-\cos(k_j\Delta x)\right] + IC\sin(k_j\Delta x)}$$

$$G = \frac{\underbrace{\left[1 - d(1-\cos k_j\Delta x)\right]}_{a} + \overbrace{IC\sin(k_j\Delta x)}^{m}}{\underbrace{1 + d(1-\cos k_j\Delta x)}_{b} + IC\sin(k_j\Delta x)}$$

$$= \frac{a + Im}{b + Im}$$

$$= \frac{(a+Im)(b-Im)}{b^2+m^2} = \frac{ab - a(Im) + b(Im) + m^2}{b^2+m^2}$$

$$= \frac{ab + Im(b-a) + m^2}{b^2+m^2}$$

$$G = \frac{ab+m^2}{(b^2+m^2)} + I\left(\frac{m(b-a)}{(b^2+m^2)}\right)$$

$$\phi_{FDE} = \tan^{-1}\left[\frac{Im_G}{Re_G}\right] = \tan^{-1}\left[\frac{m(b-a)}{(b^2+m^2)} \cdot \frac{(b^2+m^2)}{(ab+m^2)}\right]$$

$$= \tan^{-1}\left[\frac{m(b-a)}{(ab+m^2)}\right]$$

$$\therefore \quad \boxed{\phi_{FDE} = \tan^{-1}\left[\frac{C\sin(\theta)\cdot 2d(1-\cos\theta)}{1-d^2(1-2\cos\theta) - d^2\cos^2\theta + C^2\sin^2\theta}\right]}$$

$$b-a = (1 + d - d\cos\theta) - 1 + d - d\cos\theta$$
$$= 2d - 2d(\cos\theta)$$
$$= 2d(1-\cos\theta)$$

$$ab = (1-d+d\cos\theta)(1+d-d\cos\theta)$$
$$= 1 + d\cos\theta - d - d^2 + d^2\cos\theta + d\cos\theta + d^2\cos\theta - d^2\cos^2\theta$$
$$= 1 - d^2 + 2d^2\cos\theta - d^2\cos^2\theta$$
$$= 1 - d^2(1 - 2\cos\theta) - d^2\cos^2\theta$$

Dispersion errors for each component:

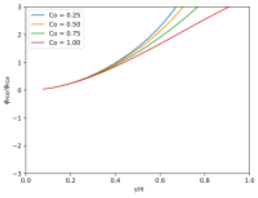
(Absolute error) $\quad \varepsilon(\phi_{PDE} - \phi_{FDE}) = \varepsilon\left(ck_j\Delta t - \tan^{-1}\left[\frac{C\sin(\theta)\cdot 2d(1-\cos\theta)}{1-d^2(1-2\cos\theta) - d^2\cos^2\theta + C^2\sin^2\theta}\right]\right)$

Relative errors:

$$\Theta = \frac{\phi_{FDE}}{\phi_{PDE}} = \frac{\tan^{-1}\left[\frac{C\sin(\theta)\cdot 2d(1-\cos\theta)}{1-d^2(1-2\cos\theta) - d^2\cos^2\theta + C^2\sin^2\theta}\right]}{ck_j\Delta t}$$

Dependent on both Δt & Δx

Plot graphs



The relative phase error is decreases with increasing Co number

Question 4

1. **What is the difference between a finite-difference (FD) method and a finite-volume (FV) method?**
   a. Finite difference uses propertues of Taylor expansion to approximate partial derivatives in a given PDE. The domain (where a PDE solution needs to be obtained) is discretized to obtain a mesh/grid where local solution at each grid point is obtained via solving system of linear equations obtained after discretization.
   b. Finite volume converts PDE's to integral forms and solves over a set of small cells that form the control volumes; hence, the domain is discretized to control volumes and the flux of interested quantities are evaluated at the volume boundaries. FVM, thus, uses cell-averaged values. FVM method is also intrinsically conservative, where the conservative laws (in - out + gen = accumulation) are held for each discrete volume.

2. **Contrast approximation of derivatives versus approximating fluxes.**
   Derivatives provide local pointwise information (I.e local changes in a function) and usually approximated using Taylor series; whereas approximating fluxes is when the governing equations are integrated over a finite control volume; flux approximations also ensure conservation.
3. **What is a cell-centered method? What is a vertex-centered method?**
   For a given mesh for FVD analysis, the mesh can be defined in these two given methods:
   a. Cell-centered is a FV discretization method that consists of selecting mesh points as the cell faces that are labeled at half-integer values ($i \pm 1/2$) and the function values are defined at the centers of these cells i.e. the unknowns are the centers of the mesh cells and the grid lines define the finite volume and surfaces
   b. Vertex centered method, in contrast, is when the unknowns are defined at the corners of the mesh. This provides a larger flexibility for definition of control volumes
4. **Define conditional and unconditional consistency. Why is conditional consistency a concern?**
   a. Consistency is defined when the solution for a given FDE matches the exact solution locally, as time and space steps tend to zero. Conditional consistency means that the consistency of the FDE does matter with the rate of dx and dt going to zero, whereas unconditional consistency means that the defined FDE is consistent regardless of how the time and space steps go to zero. Conditional consistency could be a concern because the FDE solution is very sensitive to reductions in dx and dt and it is possible to get inaccurate answers as these are refined. This makes grid sensitivity and independence study quite complex.
5. **Define conditional and unconditional stability. Define unconditionally unstable.**
   a. Stability is concerned with numerical stability of the FDE solution where a solution is stable when the stability error (exact solution of FDE without round off errors - numerical solution with roundoff errors) tends to zero (or bounded) as the number of time steps (n) increases
   b. Conditional stability means that for only for a given inequality involving a critical parameter (function of dt and/or dx) the solution is stable. Unconditional stability means that the solution is stable for all values of dt and dx. Unconditional unstable means that for no values of a given parameter is the solution stable. (I.e. unstable for all of domain since the stability condition is never satisfied)
6. **When should one use explicit methods? When should one use implicit methods?**
   a. Implicit methods are expressions where the derivatives at different mesh points appear simaltaneously to give coupled equations for solving FD method. Their advantage is in general higher order of accuracy as derivates/FDs at different mesh points are related to each other. Additionally, for a linear PDE, implicit methods generally have unconditional stability. The penalty of implicit solutions can be computational cost
   b. Explicit on other hand does not give coupled FD equations (and hence computation can be effectively parallelized).
7. **In general, which Fourier component is the most unstable?**
   a. High wavenumbers modes (high frequency) are the most unstable fourier components due to various reasons: one of them is aliasing error and higher dispersion, dissipation errors
8. **8. What is the physical meaning of the modified equation of a FD/FVE?**
   a. Modified equation is derived from taylor series expression helps identify the truncation errors, dispersion and dissipation effects. For example, the modified equation can tell us of the order of dissipation based on the lowest even order derivative term in the FDE.
9. **The Dufort-Frankel method is an explicit method. But, it is unconditionally stable and conditionally consistent. Comment**

on how to do mesh and time-step refinement.

- **a.** The method introduces time averaging term which does not match the original behavior as dt goes to 0. Therefore dx and dt need to be refined together and dt is generally on the order of dx^2 to help maintain consistency and accuracy.

10. **What is numerical dispersion**
    - **a.** Dispersion occurs when numerical method introduces phase speed errors which results in waves with different wavelentghs being propagated at incorrect speed leading to a distorted wave solution.
    - **b.** Error in the phase of the solution is defined as dispersion error
11. **Why can we use the Fourier method to analyze numerical dissipation and numerical dispersion?**
    Fourier expansions allows a quantitative estimation of errors and the decomposition separates time and space dependence of the given FDE/modified equation. The time behavior is contained in the amplitudes while the fourier modes contain the full space dependence. Therefore both dissipation and dispersion (relating to the phase) can be analyzed.
12. **Define convergence.**
    - **a.** The numerical solution must tend to the exact solution of the mathematical model (PDE) as time and space steps tend to zero
13. **Define the Lax equivalence theorem.**
    - **a.** FDEs/FVEs of well-posed, linear PDEs are convergent if the they are consistent and stable i.e. consistency and stability are necessary and sufficient conditions to show convergence under the assumptions mentioned
14. **Define conservative error.**
    - **a.** Error introduced when FD scheme fails to preserve a conserved quantity (for example entropy)
15. **Define transportive error.**
    - **a.** Occurs when a FD/FV scheme gives incorrect speed of wave leading to incorrect phase speed and wave distortion.
16. **Define aliasing error.**
    - **a.** This error results from inability of a discrete grid to effectively represent a continuous function, where for instance, the amplitude of a resolvable wavelength is changed due to choice of grid spacing. This error mainly ocurs for PDEs that are non-linear, where fourier components multiply each other and create new components that have either longer or shorter wavelengths.
17. **What is the spatial resolution of a mesh?**
    - **a.** Choosing grid spacing that allows the solution to model the physics (*i.e.* choosing appropriate dx and having enough number of grid points) but also optimizing grid points efficiency (to reduce computation time). Hence we use physics driven understanding to choose length scales that must be resolved.
18. **What is the spatial resolution of a FD/FV equation on a mesh?**
    - **a.** Discretization and replacing PDE with FDE results in additional reduction in spatial resolution
19. **What is the numerical connection between elliptic PDEs and parabolic and hyperbolic PDEs? Hint: contrast iterative space and time**
    - **a.** Elliptical PDE is solved as a global system using iterations to converge to a constant value usually via methods such as Jacobi or Gauss Seidel. These usually do not have explicit time dependence
    - **b.** Parabolic and hyperbolic PDEs are solved using iterative time step and solved using either implicit or explicit techniques

Sorry couldn't screenshot the code for some reason the image quality was getting really bad; will fix for next hw

Question 2 Code:

```
def diss_err(Co):
    # Inputs:

    dx = 0.5

    dt = dx/2
    c = Co*dx/dt
    v = 1
    d = v*dt/dx**2


    k = np.linspace(0,10,100)
    theta = k*dx/np.pi
    #x_axis = theta/np.pi

    a = 1-d*(1-np.cos(theta))
    b = 1+d*(1-np.cos(theta))
    m = Co*np.sin(theta)

    G_PDE = np.exp(-v*k*dt)
    G_FDE = np.sqrt(((a*b + m**2)**2 + m**2*(b-a)**2)/(b**2 + m**2)**2)

    err_diss = (G_PDE - G_FDE)/G_PDE
    plt.plot(theta,err_diss,label="Co = {:.2f}".format(Co))
    #

Co=np.array([0.25,0.5,0.75,1])
for i in Co:
    diss_err(i)

plt.xlabel(r" $ \gamma / \pi $",weight="bold")
plt.ylabel(r"$ |G_{PDE}| - |G_{FDE}| $");
plt.xlim([0,1])
plt.ylim([-3,3])
plt.legend()

plt.savefig("G_graph.png",dpi=800,bbox_inches = "tight")
```

Question 3 code:

```
def disp_err_rel(Co):
    # Dispersion error for each component (relative)

    dx = 0.25

    dt = dx/2
    c = Co*dx/dt
    v = 1
    d = v*dt/dx**2


    k = np.linspace(0.1,5,100)
    theta = k*np.pi*dx

    a = 1-d*(1-np.cos(theta))
    b = 1+d*(1-np.cos(theta))
    m = Co*np.sin(theta)

    phi_PDE = c*k*dt
    phi_FDE = np.arctan((Co*np.sin(theta)*2*d*(1-np.cos(theta)))/(1-d**2*(1-2*np.cos(theta))-d**2
*(np.cos(theta))**2+Co**2*(np.sin(theta))**2))

    err_phase = phi_FDE/phi_PDE
    plt.plot(theta,err_phase,label="Co = {:.2f}".format(Co))

Co=np.array([0.25,0.5,0.75,1])
for i in Co:
    disp_err_rel(i)

plt.xlabel(r" $ \gamma / \pi $",weight="bold")
plt.ylabel(r"$\phi_{FDE}/\phi_{FDE}$");
plt.xlim([0,1])
plt.ylim([-3,3])
plt.legend()

plt.savefig("phi_graph.png",dpi=800,bbox_inches = "tight")
```